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Effects of a Community-Driven Water, Sanitation, and Hygiene Intervention on Diarrhea, Child Growth, and Local Institutions

A Cluster-Randomized Controlled Trial in Rural Democratic Republic of Congo

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Abstract

Diarrhea and growth faltering in early childhood reduce survival and impair neurodevelopment. This paper assesses whether a national program in the Democratic Republic of Congo reduced diarrhea and stunting and strengthened local water and sanitation institutions. The program combined (i) funds for latrine and water upgrades, (ii) institutional strengthening activities, and (iii) behavior change campaigns. In 2018, the program was randomly assigned, after stratifying by province and cluster size, with 50 intervention and 71 control clusters. In 2022–23, 3,283 households were interviewed, at a median of 3.6 years post-intervention. The intervention had no effect on diarrhea and no effect on length-for-age Z-scores in children. Villages in the intervention group had a 0.40 higher score on the water, sanitation, and hygiene institutions index. The percentage of villages in the intervention group with an active water, sanitation, and hygiene (or just water) committee was 21 percentage points higher than the control group. Households in the intervention group were 24 percentage points more likely to report using an improved water source, 18 percentage points more likely to report using an improved sanitation facility, and reported more positive perceptions of water governance. The Democratic Republic of Congo's national rural water, sanitation, and hygiene program increased access to improved water and sanitation infrastructure, and created new water, sanitation, and hygiene institutions, all of which persisted for more than three years. However, these effects were not sufficient to reduce diarrhea or growth faltering.

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Effects of a Community-Driven Water, Sanitation, and Hygiene Intervention on Diarrhea, Child Growth, and Local Institutions: A Cluster-Randomized Controlled Trial in Rural Democratic Republic of Congo

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Introduction

The most recent estimates of the global burden of morbidity and mortality attributable to unsafe water, sanitation, and hygiene (WASH) are that 1.4 million deaths and 74 million disability-adjusted life years lost could have been prevented in 2019 [1]. People living with unsafe WASH have higher exposure to fecal-oral pathogens, resulting in enteric dysfunction, diarrheal illnesses, and, in children, growth faltering. Growth faltering, in turn, has long-term negative impacts on health, cognition, and human capital [2,3]. In 2020, 2.0 billion people did not have access to safely managed drinking water services, 3.6 billion did not have access to safely managed sanitation services, and 2.3 billion did not have access to handwashing facilities with soap and water at home [4,5]. While access has been increasing, progress will need to accelerate by three-to-six-fold to meet the Sustainable Development Goals for 2030 [6]. The challenge of increasing access is particularly acute for people living in or near armed conflict – one in six people worldwide – both through the direct effects of conflict and because violence and insecurity impede collective action to provide public goods like WASH [7,8]. In the Democratic Republic of Congo (DRC), as of 2020, 48 million people still lacked basic drinking water services, 11 million people still practiced open defecation, and 72 million people still lacked basic hygiene services [9].

To increase access to safe WASH, governments and donors have increasingly turned to community-led approaches. While WASH experts called for greater community participation for over 30 years [10], the sector did not fully embrace this approach until the late 1990s. Beginning with community-led total sanitation in Bangladesh in 1999, community-led WASH programs have been implemented in at least 60 countries, and 15 countries have incorporated them into national policy [11,12].

Despite this broad adoption, the health effects of community-led WASH interventions – and of many WASH interventions in general – remain poorly understood. The accumulation of evidence has accelerated in recent years, but the length of the causal chain from the intervention to the outcome, and the vast design space for WASH interventions (which can incorporate behavior change campaigns, infrastructure, institutions, and/or new technologies), means that many fundamental questions remain unanswered. A meta-analysis of 13 randomized WASH trials found no effect on child length-for-age, but a meta-analysis of 124 WASH studies (randomized and observational) found a protective effect against diarrhea [11,13–24]. There is a great deal of heterogeneity in effect size across studies, likely due to variation in intervention components and intensity, and to the influence of contextual factors such as baseline exposure to fecal matter.

To our knowledge, this is the first trial of an intervention that combines the creation of new institutions, funding for new or improved infrastructure, and a behavior change campaign, all within a locally-led process of targeting and implementation. We study this complex intervention as it is implemented at scale, providing a realistic estimate of effectiveness for policy makers in similar contexts. Our follow-up period is unusually long (3.6 years), enabling us to address the question of sustainability. We also provide evidence in a conflict-affected setting, where WASH interventions have rarely been evaluated using experimental designs.

The intervention is the ‘healthy villages’ component of the DRC’s Healthy Villages & Schools program, co-led by the Ministries of Public Health, and of Primary, Secondary, and Professional Education, with support from the United Nations Children’s Fund (UNICEF). Since 2008, nearly 9 million people in almost 11,000 villages have been reached with WASH services through the program. Healthy Villages & Schools was the largest WASH program implemented by UNICEF globally and comprised 90% of total external funding committed to rural WASH in the DRC from 2005 to 2020 [25]. Our goal was to estimate the effect of Healthy Villages & Schools on diarrhea prevalence, child length-for-age, and WASH institutions.

Methods

Study design and participants

The intervention of interest is a DRC government-run program that began several years before our study. For our study, the government agreed to randomly assign the next phase of the program (i.e. the next group of villages to receive the intervention). In 2018, we randomly assigned groups of villages to intervention or control (details below). Since we did not collect any data prior to randomization, we did not yet register the study. We collected the first round of data in late 2019, about 5 months after the intervention was implemented (implementation took about one year in each group of villages) [26].

We pre-registered the analysis of that first round of data collection (5 month follow-up), before any of the authors saw the data, in the American Economics Association (AEA) Registry (AEARCTR-0004648) (https://www.socialscienceregistry.org/trials/4648).

In Feb 2021, we registered plans for additional data collection in the Pan-African Clinical Trials Registry (PACTR202102616421588; https://pactr.samrc.ac.za/TrialDisplay.aspx?TrialID=14670). In April 2023, before the PIs had seen any data for the 3.6-year follow-up (i.e., the data for the current manuscript), we updated the registration with our primary and secondary outcomes. In April 2023, we also posted our pre-analysis plan for the 3.6 year follow up on the AEA Registry (see "Three Year Follow up Pre-Analysis Plan" at https://www.socialscienceregistry.org/trials/4648). All planned studies from this project are now registered and any future work will be registered prospectively.

This study is reported as per CONSORT guideline (S1 Text).

We worked with intervention implementers to design a cluster-randomized trial in rural villages in five DRC provinces: Kongo Central, Kasai, Kasai Central, North Kivu, and South Kivu. The implementers identified 403 candidate villages in which the intervention could be launched during the study period, based on the established criteria for the intervention: that the village was located in a secure and accessible Health Area that was not already served by the WASH Consortium, the Health Area staff were dynamic and interested in participating, and there was a problem of diarrhea, cholera, and/or malnutrition. Among these villages, 34 already had program activities in process before research activities began, leaving 369 eligible villages.

To avoid spillover effects from treatment villages to control villages, we grouped those villages into clusters. We considered any villages within 2.5 km of each other (using Euclidean distance between village centroids) to be part of the same cluster. Therefore, all clusters have at least 2.5 km between them. We relax this rule in South Kivu, where density is greater, and use a minimum distance of 1km. In total, this resulted in 124 clusters. North Kivu had only three clusters (covering 30 villages); as a result, it was not logistically feasible to include these villages in the trial. That left 121 clusters (339 villages) in four provinces.

Each village in the intervention clusters received the intervention, as described above. Villages in control clusters did not receive any intervention. Data collection procedures were identical in the two groups.

The study protocol was approved by the Institutional Ethics Committee of the *Institut Superieur des Techniques Médicales de Bukavu* (DRC) (#001/2019 & #008/2022) and by Solutions IRB (USA) (#2019/10/20). With direction from the study investigators, Innovative Hub for Research in Africa (IHfRA) was responsible for data collection. No study data was collected by intervention implementers.

All residents of the targeted villages who had lived in the village for at least four years (i.e., moved to the village prior to the intervention) were eligible to participate in the study. Two groups of households were interviewed: households that were interviewed at 5 months post-intervention (4 per village) and households that had not been previously interviewed (6 per village). Both groups were randomly selected (at their respective entries into the study) as follows: from the center of the village, interviewers went in opposite directions to the nth household, where n was a randomly selected number between 1 and 20. We interviewed the head woman of the household. We also asked to measure the height and weight of the youngest child aged 2-5 years, or, if none, the oldest child aged 0-2 years. Adult participants provided informed consent verbally, which was recorded electronically.

Randomization and masking

A total of 339 villages in 121 clusters were eligible for randomization. In all provinces except Kasai Central, each cluster was given equal probability of being selected for treatment or control. In Kasai Central, due to budget constraints, we increased the probability of being selected into the control group to 75%, to reflect the fact that only 16 out of 81 villages could receive the intervention. Thus the allocation probability for the intervention group was 25%.

We block randomized by province and number-of-villages-per-cluster (12 blocks total; see S1 Table). Since randomization was based on clusters but the implementing organization's operational targets were based on villages, it was not possible to force the randomization to select the exact number of villages targeted without introducing potential bias. Instead, we compared the number of target villages per province to the number of treatment villages selected after randomization. In cases where the number of intervention villages was larger than the operational target, we randomly dropped an equal number of intervention villages from the largest control and intervention clusters until operational targets were met. We dropped 2 villages in Kongo Central and 4 villages in Kasai. We also dropped one control village in Kasai due to a coding error. This left 146 villages in 50 clusters in the intervention group, and 186 villages in 71 clusters in the control group. S1 Table shows how intervention and control villages and clusters are distributed across provinces. Randomization was done by the research team in Stata.

Due to the participatory and visible nature of the intervention, neither participants nor data collectors were masked to treatment status. However, data collectors did not participate in intervention implementation and were employed by a separate, independent organization.

Procedures

The intervention, “Healthy Villages & Schools”, was developed by the DRC government and UNICEF. We focused on the village rather than the school component. This program mobilizes communities to become a “Healthy Village” with 3-6 months of support from government health officials and local NGOs, including approximately $2,000 of financing for new or improved water infrastructure, $2,000 for new or improved sanitation infrastructure, and $3,000 for personnel costs, per village. The mean village size in the intervention group was 456 people (median 400; IQR 502). The seven norms to become a Healthy Village are:

1. There is a dynamic village WASH committee.
2. At least 80% of the population has access to safe drinking water.
3. At least 80% of households use a hygienic latrine.
4. At least 80% of households dispose of their household waste hygienically.
5. At least 60% of the population washes their hands before eating and after going to the latrine.
6. At least 70% of the population is aware of fecal-oral disease transmission and how to prevent this.
7. The village is cleaned at least once a month.

The program is implemented in nine steps (Table 1) [27].

Table 1. The Healthy Village and Schools Program’s Nine Steps

Step	Description
0	The community learns about the program and collectively decides to adopt it before submitting a formal request to the relevant Health Zone. (A Health Zone is a geographic unit of the Congolese health system that contains roughly ten Health Areas and 100,000 residents, run by a Chief Medical Officer (CMO)). Program protocols state that the entire community should be involved in the decision to participate.
1	A statement of agreement between the community and the Health Zone is signed.
2	Health Zone officials survey 19 households on knowledge, attitudes, and practices (KAP). The community self-evaluates on eight practices, including handwashing, water use, and sanitation.
3	The community spends about 11 hours over five days creating calendars and maps, visiting water points, classifying hygienic practices as healthy or unhealthy, discussing fecal-oral disease transmission, calculating medical costs, and assessing which individuals and organizations influence sanitation and hygiene in the community. This includes 1.5-2 hours in a facilitated activity around the question, “What are the hygiene practices that we want to change in our village?”

4	The Health Zone provides training for 20 volunteers on maintenance of latrines, water supply systems, and sanitation, conflict management, and petty cash management. The community elects a village WASH committee.
5	The community spends ten hours over three days describing a community vision, analyzing the barriers to reducing diarrheal diseases, choosing improvements to drinking water, sanitation, and hygiene, and formulating an action plan. The community is asked to identify practical, low-cost solutions with a minimum of outside assistance. New infrastructure is evaluated in terms of accessibility, technical feasibility, and technical capacity.
6	The community builds new infrastructure over 90-180 days, supported by project funds. Key messages about sanitation and hygiene are discussed during sensibilization meetings or during visits to families by the WASH committee, community health workers, or other volunteers. Health Zone staff are expected to visit the community monthly during this time; Health Area staff weekly.
7	The community self-evaluates again, to measure progress since Step 2. The Health Zone conducts additional KAP surveys and hosts three hours of meetings to assess the findings and make a plan to maintain progress.
8	The CMO spends one day in the community to assess whether or not the community has completed its action plan and achieved the seven norms. If they have, a certification ceremony is held. The CMO and the village WASH committee develop a Community Action Plan for Maintenance so that the changes achieved through the program can be sustained over time.

The IHfRA data collection team used electronic tablets and transmitted data to a cloud-based server, allowing the research team to conduct quality control measures in real-time, checking for consistency and errors. Additionally, IHfRA randomly selected 15% of villages for a second round of interviews, by different interviewers, with a shorter questionnaire, to check consistency across key variables. Separately, children from two households per village had their height and weight re-measured by an IHfRA supervisor, as a quality check.

Outcomes

Primary outcomes were caregiver-reported diarrhea in the last seven days among all children who were under 5 years old at the time of the survey, length-for-age Z score for a randomly selected child in each household, and a WASH governance index. If the household had one child between age 2 and 5, we measured the length and weight of that child. If the household had more than one child between age 2 and 5, we randomly selected one child. If the household had no children between ages 2 and 5, but one or more children between ages 0 and 2, we measured length and weight of one of those children (randomly selected). Salter scale (Model 235 6S) and wall-mounted measuring rods (portable baby/child length/height measuring system) were used.

The WASH governance index combined questions about the presence of a water committee, frequency of committee meetings, WASH expenses (excluding maintenance), presence of a maintenance plan, whether the committee tracks health conditions in the community, and whether it tracks hygiene and sanitation.

Secondary outcomes were access to improved water and sanitation facilities, water quality at water points and in homes, hygiene knowledge and behaviors, observed handwashing, perceptions of WASH governance, children's school absenteeism, child weight-for-age Z score, and child weight-for-height Z score.

Structured observation of handwashing was done in four households per village. We first attempted to observe the four households that were interviewed at the five-month follow-up; if any were unavailable or unwilling, we randomly selected from the six new households to replace them. A research assistant spent two hours in each of these households, recording if handwashing occurred at critical junctures such as before preparing food (see S8 Table for full list of junctures). This took place before the interview, to minimize Hawthorne effects.

Access to improved water and sanitation was self-reported, i.e. respondents reported whether their main water source is improved or not according to the Joint Monitoring Program standard definitions (e.g. boreholes are considered improved, while unprotected springs or surface water sources are not). Cost paid by households for water and time spent collecting water were also self-reported.

To measure water quality, we tested samples collected (i) at each of the water points used by members of each village, and (ii) at household water storage containers in six randomly selected households per village, on average. Testing was done concurrently with the household interviews. We used the Aquagenx Compartment Bag Test E. Coli +Total Coliform Most Probable Number (MPN) Kit. This measured the MPN of fecal indicator bacteria [28].

To measure subjective performance of local WASH institutions, we used survey questions about: fairness of selection of water governance entity, perception of fair treatment, confidence in the entity's management of money, confidence in the entity's response to infrastructure breakdowns, confidence in management, and overall satisfaction. The data collection team also directly observed water point functionality. Length of breakdowns was reported via a water committee and village leader survey.

Statistical analyses

The number of village clusters in the study was determined by the program budget and proximity of villages to one another (details above). Sample size calculations were used to determine how many households in each village should be interviewed. Based on the primary outcome of diarrhea prevalence, with a minimum detectable effect of 8 percentage points (pp), 32%

prevalence in the control group (based on the 5-month follow-up results), intracluster correlation of 0.09 (based on 5-month follow-up), and 1.3 children under 5 years per household (based on 5-month follow-up), we required 10 households per village.

We estimated intervention effects according to random assignment (intention to treat), irrespective of adherence to the intervention.

For both primary and secondary outcomes, whether binary or continuous, we fitted linear models with a binary variable indicating whether the participant was in a treatment or control cluster [29]. Because randomization was stratified by province and number-of-villages-per-cluster, we included binary variables (fixed effects) for each stratum (n=12) in the model. We also included gender and age (month) indicator variables for all child health outcomes. We clustered standard errors at the cluster (i.e. group of villages) level.

For the primary outcome of WASH institutions, and secondary outcomes consisting of multiple measures, we calculated a summary index to avoid over-rejection of the null hypothesis due to multiple inferences. We rescaled each outcome so that higher values implied better outcomes, and averaged standardized values relative to the control group. Treatment effects were estimated as the difference in the summary index between treatment and control groups, such that treatment effects are expressed in standard deviation units relative to the control group.

We pre-registered two analyses restricted by subgroup: by province (for all three primary outcomes), and by gender (for diarrhea and length-for-age). We test for interaction on the additive scale, using interaction terms in linear models.

Statistical analyses were conducted in Stata version 16.0.

Role of the funding source

This study was funded by the UK Foreign, Commonwealth, and Development Office (FCDO) (https://www.gov.uk/government/organisations/foreign-commonwealth-development-office), via Amendment No. 3 to the Supplemental Arrangement with the World Bank regarding Multi-Donor Trust Fund for Impact Evaluation to Development Impact (TF072617, parallel to TF072161). AC, KC, and EM were staff at Development Impact at that time. The Healthy Villages & Schools program was a DRC government national program funded by UK's FCDO and implemented with UNICEF's support. The funder and implementing partners provided inputs at the design stage to ensure the study addressed policy and program priorities of importance to them. The funders had no role in data collection and analysis, decision to publish, or preparation of the manuscript.

Results

Of the 1,312 respondents in 328 villages interviewed for the 5-month follow-up, 1,133 (86%) in 328 villages were re-interviewed at the 3.6-year follow-up, between November 24, 2022 and February 10, 2023 (Figure 1). We also reached two villages that were not accessible during 5-month follow-up and lost one village that was surveyed at 3.6-year follow-up due to conflicts. In 39 households with a 5-month follow-up interview, a new respondent was interviewed at 3.6

years, and 140 households (11%) were replaced between 5-month and 3.6-year follow-ups. Additionally, in each village at the 3.6-year follow-up, six never-previously-interviewed households were randomly selected, conditional on having lived in the village for at least four years, yielding 1,970 interviews (in four villages, only five households were reached). Thus, at 3.6 years, we interviewed a total of 3,283 households. Of those households, 75% (2,466 out of 3,283) had at least one child eligible for a caregiver's reports of diarrhea, and 72% (2,374 out of 3,283) had at least one child eligible for length and weight measurement. The primary outcome of WASH institutions was measured in 329 villages. In the intervention group, the median time since the completion of Healthy Villages Step 6 (construction of infrastructure) was 3.6 years (IQR = 3.4 to 3.7).

At 3.6 years, respondents in intervention and control groups were similar with regard to characteristics unlikely to be affected by the intervention, such as marital status, educational attainment, age, religion, household size, and home construction materials (Table 2).

Table 2. Household and respondent characteristics by intervention group, at 3.6-year follow-up

	Control			Intervention				
Outcomes	n	Mean	SD	n	Mean	SD	Adj. Diff.	p-value
Household has improved roof	1843	0.42	0.49	1436	0.47	0.50	-0.01	0.81
Household has improved wall	1845	0.01	0.09	1438	0.01	0.12	0.01	0.18
Household has improved floor	1816	0.04	0.19	1430	0.08	0.27	0.03	0.11
Household size	1845	7.20	2.86	1438	7.18	2.93	0.03	0.80
Respondent identifies as Catholic	1845	0.18	0.38	1438	0.19	0.40	0.02	0.54
Respondent identifies as Protestant	1845	0.31	0.46	1438	0.32	0.47	-0.06	0.05
Respondent identifies with other religion	1845	0.02	0.15	1438	0.03	0.18	0.01	0.26
Respondent age	1845	40	13.39	1438	40	13.07	0.72	0.25
Respondent has completed primary school	1845	0.31	0.46	1438	0.34	0.48	-0.02	0.50
Respondent has completed secondary school	1845	0.06	0.23	1438	0.07	0.25	0.00	0.78
Respondent is married or cohabitating	1845	0.83	0.37	1438	0.82	0.38	-0.01	0.49

Adj. diff. = adjusted difference between intervention group and control group, estimated with models that include controls for randomization blocks based on province and number of villages per cluster, and standard errors clustered by cluster. There were 121 clusters in total. Improved roof=1 if roof is finished roofing (i.e., metal, wood, calamine/cement fiber ceramic tiles, cement or roofing shingles); improved walls=1 if walls are 'finished walls'; improved floor=1 if floor is 'finished floor'. All variables are binary except 'HH size' and 'respondent age'; for these binary variables, the mean represents the proportion of respondents who are in the listed category.

In the intervention group, 96% of villages reported that they created a community action plan and prioritized actions to improve WASH (as instructed by the program); 86% reported that they had implemented that plan.

The intervention had no effect on diarrhea (adjusted mean difference -0.01 [95% -0.05– 0.03]) (Table 3). Diarrhea prevalence was high overall, at 38% in the treatment group and 42% in the control group. The intra-cluster coefficient (ICC) for diarrhea in the control group was 0.05; in the intervention group, 0.07.

Table 3. Intervention effects on primary outcomes: diarrhea, length-for-age, and WASH institutions

	Control				Intervention						
Outcomes	n	Prevalence/Mean	SD	ICC	n	Prevalence/Mean	SD	ICC	ITT	CI 95%	
Diarrhea prevalence	2310	42%		0.05	1762	38%		0.07	-0.01	-0.05	0.03
Length-for-age Z-score	1223	-2.18	1.60	0.04	919	-2.20	1.59	0.06	-0.01	-0.15	0.12
WASH institutions index	185	0.00	1.00	0.47	144	0.46	0.75	0.11	0.40	0.16	0.65

ITT = intention-to-treat effect estimate; ICC =intracluster correlation; HH = household. Effects are estimated with models that include controls for randomization blocks based on province and number of villages per cluster. There were 121 clusters in total. The WASH institutions index was calculated by rescaling each variable in the index (eg, presence of WASH committee) so that higher values imply better outcomes, then standardizing relative to the control group, following Kling *et al.* Effects are in standard deviation units.

The intervention had no effect on length-for-age Z-scores in children (adjusted mean difference -0.01 [95% CI -0.15–0.12]). In the control group, the mean length-for-age Z-score was -2.18 (1.60 SD) (Figure 2). The ICC for length-for-age Z-score in the control group was 0.04; in the intervention group, 0.06.

Villages in the intervention group had a 0.40 higher score on the WASH institutions index (95% CI 0.16–0.65). The percentage of villages in the intervention group with an active water, sanitation, and hygiene (or just water) committee was 21 pp higher than the control group. The ICC for the WASH institutions index in the control group was 0.47; in the intervention group, 0.11.

Households in the intervention group were 24 pp (95% CI 12–36) more likely to report using an improved water source, 18 pp (95% CI 10–25) more likely to report using an improved sanitation facility, and reported more positive perceptions of water governance (adjusted difference 0.19 SD [95% CI 0.04–0.34]) (Table 3). The more positive perceptions of water governance were driven by higher reported satisfaction with water access (0.56 points higher (95% CI 0.27–0.85) on a 1-5 scale). Intervention group households were also 9 pp more likely to report paying for water (95% CI 0-19). Conditional on paying for water, there was no difference in the amount paid between the intervention and control groups. The intervention had no effect on time spent collecting water.

Intervention group water points were 11 pp less likely to be currently functional (95% CI -0.18– -0.05); 97% of water points in control villages and 85% of water points in intervention villages were functioning.

Intervention group respondents scored 0.23 higher (95% CI 0.12–34) on the index of self-reported hygiene & behavior index. This was driven by several measures. The intervention group was 3 pp more likely to report treating their water (95% CI 1–5), and 9 pp more likely to report handwashing with soap or ash at least once in the previous day (95% CI 6-13). The intervention group also scored 0.43 points higher (95% CI 0.19–0.67) on the handwashing knowledge scale (range 0-10). However, in structured observations of handwashing behavior, there was no difference between intervention and control households in measures of any observed handwashing or handwashing with soap or ash.

Water samples from intervention village water points showed a small but statistically significant improvement in thermotolerant coliforms per 100mL compared to control village samples (-0.17 adjusted mean difference in log10(MPN); 95% CI -0.32– -0.02). Overall water quality was low, even from improved sources. Among unimproved sources, 86% of samples had coliform levels over 100 per 100mL, 'very high risk' according to WHO standards [30]; among improved sources, 60% of samples had coliform levels over 100 per 100mL. Water samples from intervention household water containers showed no difference in thermotolerant coliforms per 100mL compared to control household samples.

The intervention had no effect on the psychological well-being index, the life satisfaction & self-esteem index, or school attendance.

In the prespecified subgroup analysis of primary outcomes by province, we find that the intervention reduced diarrhea in one of four provinces (Kongo Central), reduced length-for-age in one province (Kasai Central), and increased the WASH institutions index in two provinces (Kasai and Kasai Central) (see S3 Table). We also used interaction terms in linear models to test for effect modification on the additive scale (S5 Table). Of the nine coefficients (three provinces, leaving out a reference, and three primary outcomes), two were statistically significant: in Kasai Central province, the results suggest that the intervention increased diarrhea prevalence relative to the treatment effect in the reference province (Kongo Central); in Kasai province, the results suggest that the intervention increased length-for-age. For both diarrhea and length-for-age, the Wald test rejects the null hypothesis that all province-by-intervention coefficients are zero at the 0.05 level. For the WASH institutions index, the Wald test fails to reject the null.

In the prespecified subgroup analysis of diarrhea and length-for-age by child sex, we find no difference in intervention effects by sex (see S4 Table). In linear models with interaction terms for intervention-by-sex, the coefficients are not statistically significant (S7 Table).

Discussion

We tested the effects of the national community-led rural WASH program in the DRC on child length-for-age, diarrhea, and WASH institutions. The program improved community WASH institutions, with intervention villages more likely to have a WASH committee, and for this institution to actively monitor community health conditions. Intervention villages also had greater access to improved water and sanitation infrastructure as a result of the program. However, we cannot reject the null hypothesis that the intervention did not have any effect on child length-for-age or diarrhea.

The finding of no effect on length-for-age is consistent with a recent meta-analysis of 11 WASH trials with length-for-age as a primary outcome that found an adjusted mean difference in Z-score between intervention and control of 0.00 (95% CI -0.03–0.04) [13]. These trial results stand in contrast to many observational studies finding that WASH protects against growth faltering [31]. This suggests that the observational results may be confounded by other household or community characteristics.

We measured fecal indicator bacteria in water sources and household water containers. We found no difference between the intervention and control group household water quality, despite the fact that intervention households were 24 pp more likely to use an improved water source. This is likely due to the fact that improved water sources in our study had low-quality water, consistent with evidence from DRC [25] and elsewhere [33]. It may also be linked to recontamination of water between collection and its ultimate use in the household. Overall, intervention water points had only slightly lower levels of fecal indicator bacteria than control water points; log10MPN/100mL was 0.17 lower in intervention water points (95% CI -0.32– -0.02), with a mean of 1.71 in the control group. This is consistent with a meta-analysis of five WASH trials that found only a 6% reduction in prevalence of enteropathogens in environmental samples [32].

In addition, two-thirds of our respondents reported spending at least one day per week working on an agricultural plot (modal response = 4 days). Of those who did, 95% report open defecation while in the field, and 91% report drinking from surface water or unprotected springs. This highlights the challenge of delivering safe and comprehensive WASH services in some agricultural settings.

Our finding of no effect on diarrhea contrasts with a meta-regression (conducted as part of a systematic review) which found use of an off-site improved water source reduces the relative risk of diarrhea by 19% compared to unimproved water [14]. The same analysis found that basic sanitation without sewer connection lowered the relative risk of diarrhea by 21% relative to unimproved or limited sanitation, which also contrasts with our results, while hygiene interventions reduced the relative risk of diarrhea by 30%. However, another review found that effective handwashing promotion typically requires daily to fortnightly contact between the promoter and participant [31]. It is possible that our intervention achieved that frequency of contact during the most intensive 90-180 days of implementation, but also likely that effects would have faded out by our measurements over three years later.

We found no effect of the intervention on handwashing with soap or ash during structured observations of study participants by our research team. However, participants in the intervention group were 9 pp more likely to report washing with soap or ash the previous day. This underscores the limitations of self-reported data, particularly when the socially desirable outcome is likely to be known and salient.

Sustainability is a widely-recognized challenge for WASH interventions. At five months post-intervention, Healthy Villages and Schools increased access to improved water sources and improved sanitation facilities[26]. Notably, these improvements largely persisted to 3.6 years post-intervention, as did the improvements in WASH institutions. Yet these improvements did not result in any measurable effect on diarrhea or growth faltering. This highlights that it is crucial to measure health outcomes directly and not assume that better inputs are sufficient to yield improvements.

This study has several limitations. First, the trial had incomplete adherence: 86% of villages in the intervention group reported that they had implemented the community action plan to address WASH challenges (e.g. by building new infrastructure) by the 3.6-year follow-up. However, this is a realistic level of adherence for a government-implemented program. Indeed, given that many study villages were conflict-affected, the take-up rate was substantial. Second, we have no baseline measures. Although the randomized design means that such measures are not required for unbiased estimates of treatment effects, there are theoretical challenges; for example, if permanent migration out of study villages was affected by the intervention, then our estimates may be biased. However, at 3.6 years post-intervention we were able to re-interview 88% of the households interviewed at 5 months post-intervention, suggesting that migration was relatively rare in our study population. We also restricted households that were newly recruited at 3.6 years to those who had lived in their current residence for at least four years. Third, the outcomes that are self-reported may suffer from reactivity or social-desirability bias.

Our results reinforce calls for more ambitious attempts to improve WASH services to reduce stunting and diarrhea, such as "transformative WASH." [34] Proponents of transformative WASH argue that some or all of the following may be necessary to produce significant health gains: high community coverage of improved sanitation facilities; living environments free from animal feces; continuous, convenient access to clean water; new approaches to behavior change; or new technologies to deliver WASH services. [31] Others go further and argue for transformative housing, with connections to water and sanitation networks. [12] These critiques are relevant in our setting, given the multiple potential sources of contamination, the low quality of even "improved sources", and the high burden of disease. Business as usual is not enough.

Data sharing statement

Individual-level, de-identified data from this study and code to reproduce all results are publicly available in the World Bank micro-data catalogue here:
https://reproducibility.worldbank.org/index.php/catalog/239.

Figure titles and legends

Fig 1. Trial Profile.
HH = household

Fig 2. Distribution of length-for-age Z-scores, intervention and control groups
Length-for-age Z-scores for children aged 0-5 years, in the intervention group (n=919) and the control group (n=1223).

Supporting information

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Figures

Figure 1. Trial profile

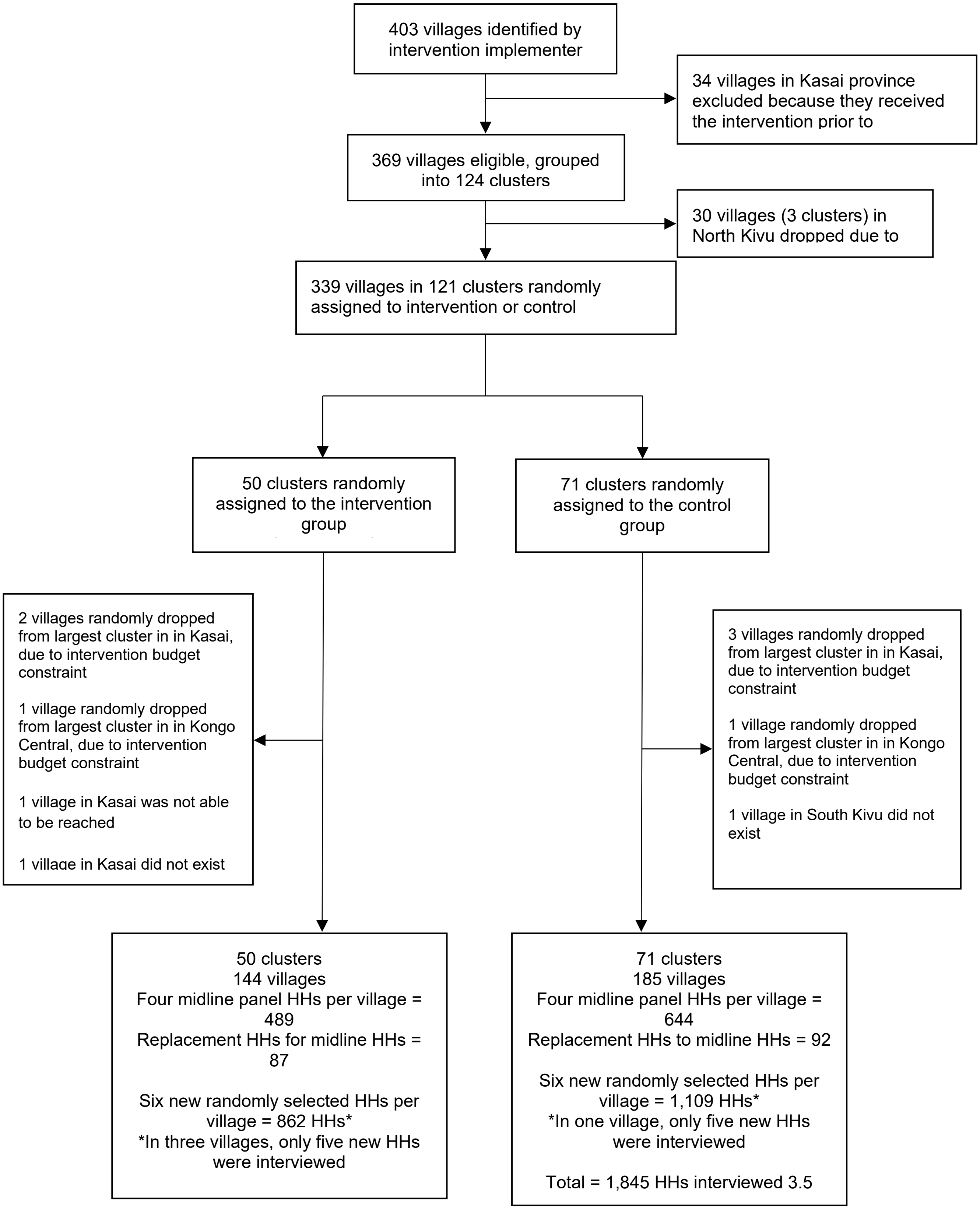
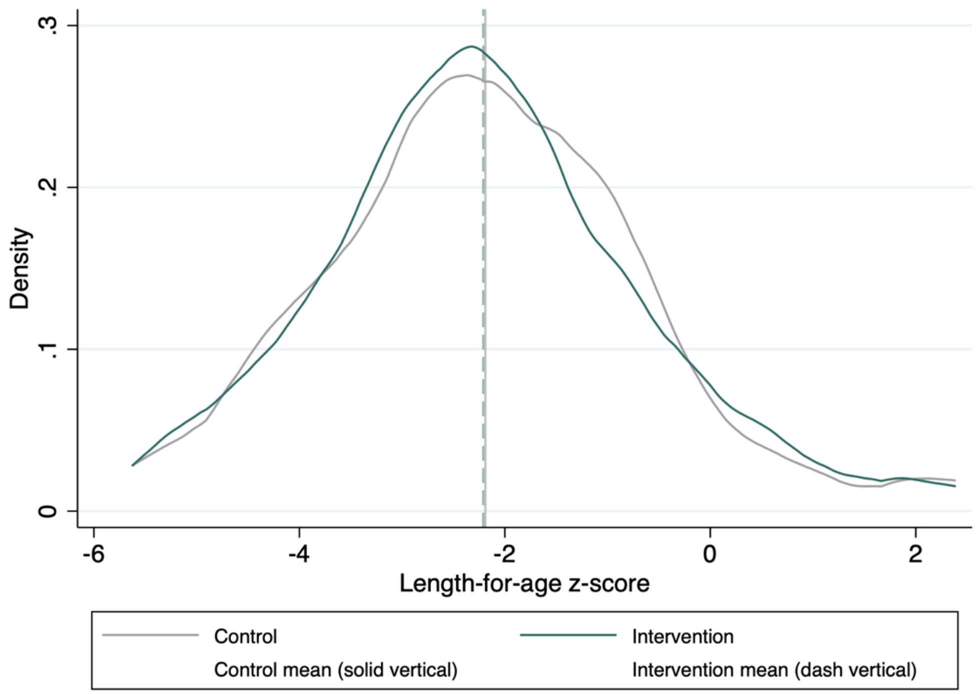


Figure 2. Distribution of length-for-age Z-scores, intervention and control groups



Supporting Information

S1 Table. Randomization strata

Province	Stratum	Villages-per-cluster	Total villages in stratum	Intervention villages in stratum	Control villages in stratum	Total clusters in stratum	Intervention clusters in stratum	Control clusters in stratum
Kongo Central	1	1-2	15	8	7	12	6	6
	2	3-4	13	6	7	4	2	2
	3	5,7	12	7	5	2	1	1
Kasai	1	1-2	37	19	18	26	13	13
	2	3-5	39	21	18	11	6	5
	3	10,12	22	12	10	2	1	1
Kasai Central	1	1-2	34	8	26	25	6	19
	2	3-4	34	8	26	10	2	8
	3	6-7	13	0	13	2	0	2
South Kivu	1	1-2	20	9	11	13	6	7
	2	4-8,10	74	39	35	12	6	6
	3	12,14	26	12	14	2	1	1
Total			339	149	190	121	50	71

S2 Table. Intervention effects on WASH institutions index and index sub-components

	Control			Intervention				CI 95%	
Outcomes	n	Mean	SD	n	Mean	SD	ITT	Lower Bound	Upper Bound
WASH institutions index	185	0.00	1.00	144	0.46	0.75	0.40	0.16	0.65
Committee (y/n)	185	0.70	0.46	144	0.97	0.16	0.21	0.10	0.32
Committee mtg freq*	88	2.91	1.49	104	2.62	1.61	-0.29	-0.79	0.20
WASH expenditures (CDF per month, IHS)	130	2.12	4.07	140	3.28	4.58	1.14	0.07	2.20
Track health (y/n)	185	0.69	0.46	144	0.84	0.37	0.16	0.07	0.25
Track sanitation (y/n)	185	0.74	0.44	144	0.78	0.41	0.05	-0.06	0.15

ITT = intention-to-treat effect estimate. CDF = Congolese francs. IHS = inverse hyperbolic spline. Effects are estimated with models that include controls for randomization blocks based on province and number of villages per cluster. There were 121 clusters in total. The WASH institutions index was calculated by rescaling each variable in the index (e.g., presence of WASH committee) so that higher values imply better outcomes, then standardizing relative to the control group, following Kling *et al*. Effects are in standard deviation units. The index values range from -2.35 to 2.12.

*Committee meeting frequency is coded 1-6, where 1=weekly, 2=Fortnightly, 3=Monthly, 4=Every 3 months, 5=Every 6 or more months, 6=No regular schedule, based on needs

S3 Table. Intervention effects on all secondary outcomes, including index sub-components

	Control			Intervention				CI 95%	
Outcomes	n	Prevalence/ Mean	SD	N	Prevalence/ Mean	SD	ITT	Lower	Upper
Improved water source	1845	43%		1437	74%		0.24	0.12	0.36
Improved sanitation	1843	14%		1437	34%		0.18	0.10	0.25
Water governance perception index	1845	0.00	1.00	1438	0.22	0.88	0.19	0.04	0.34
Committee selected fairly	600	1.76	0.81	1119	1.82	0.76	0.06	-0.04	0.17
Committee treats community fairly	621	4.15	1.31	1158	4.07	1.31	-0.09	-0.27	0.09
Committee manages money well	635	3.25	1.12	721	3.31	1.14	0.05	-0.13	0.23
Comm responds breakdown well	972	3.69	1.18	1227	3.67	1.23	0.01	-0.13	0.15
Confidence committee will solve reported issue	1009	3.10	1.05	1227	3.12	1.03	0.02	-0.13	0.17
Satisfaction with water access (1-5)	1845	2.76	1.49	1438	3.38	1.43	0.56	0.27	0.85
Time spent collecting water (IHS)	1845	4.17	2.07	1438	4.04	2.05	0.03	-0.20	0.27
HHs pay for water (Dummy)	1845	10%		1438	23%		0.09	0.00	0.19
Water use expenditure (weekly)	137	153	217	303	254	351	114	-16	243
Hygiene and behaviour index	1845	0.00	1.00	1438	0.30	1.07	0.23	0.12	0.34
Handwashing score (0-10)	1845	2.79	2.22	1438	3.41	2.16	0.43	0.19	0.67
Open defecation (%)	1843	32%		1437	27%		-0.03	-0.08	0.01
Self-reported handwashing with soap/ash (%)	1843	67%		1437	82%		0.09	0.06	0.13
Frequency of latrine cleaning over past 2 weeks	761	4.89	4.26	648	5.00	4.06	0.25	-0.32	0.82
Water in pot is clean and covered (%)	1669	57%		1273	57%		0.02	-0.03	0.07
Water treated for consumption, any method (%)	1669	3%		1273	7%		0.03	0.01	0.05
Handwashing action	616	15%		489	17%		0.00	-0.04	0.03
Handwashing with soap/ash	616	2%		489	4%		0.01	-0.01	0.03
Water point has water	380	97%		266	85%		-0.11	-0.18	-0.05
WP FIB MPN/100mL	366	79.31	37.92	221	72.41	42.56	-9.46	-18.85	-0.06
WP FIB log10 MPN/100mL	366	1.71	0.60	221	1.58	0.70	-0.17	-0.32	-0.02
HH FIB MPN/100mL	1098	82.27	35.05	851	79.44	37.71	-3.87	-9.59	1.86
HH FIB log10 MPN/100mL	1098	1.76	0.53	851	1.72	0.58	-0.06	-0.15	0.04
Weight for age	1223	-1.17	1.16	919	-1.14	1.16	0.06	-0.04	0.15
Weight for length	1224	0.14	1.07	918	0.22	1.03	0.10	-0.02	0.22
Life satisfaction & self-esteem index	1843	0.00	1.00	1435	0.03	0.98	0.01	-0.12	0.14
Life satisfaction (WVS)	1843	4.87	3.20	1435	5.04	2.89	0.20	-0.10	0.49
Feel I am person of worth (Rosenberg)	1843	2.25	0.90	1435	2.19	0.90	-0.03	-0.13	0.07
Feel that I have good qualities (Rosenberg)	1843	2.42	0.74	1435	2.36	0.75	-0.02	-0.10	0.05
Inclined to feel I am a failure (Rosenberg)	1843	1.43	1.11	1435	1.49	1.05	-0.04	-0.16	0.08
Able to do things as well as other people (Rosenberg)	1843	2.44	0.75	1435	2.40	0.76	0.01	-0.06	0.08
Feel have not much to be proud of (Rosenberg)	1843	0.98	0.98	1435	0.99	0.92	-0.05	-0.14	0.04
Take a positive attitude towards self (Rosenberg)	1843	2.35	0.78	1435	2.37	0.74	0.08	-0.01	0.17
I am satisfied with myself (Rosenberg)	1843	2.03	1.02	1435	2.12	0.96	0.16	0.07	0.26

Wish could have more respect for myself (Rosenberg)	1843	0.42	0.60	1435	0.40	0.57	-0.07	-0.13	0.00
Certainly feel useless at times (Rosenberg)	1843	1.33	1.10	1435	1.40	1.07	-0.03	-0.16	0.10
At times think I am no good at all (Rosenberg)	1843	1.32	1.09	1435	1.43	1.07	0.00	-0.12	0.12
Psychological well-being index	1843	0.00	1.00	1435	0.05	0.99	0.06	-0.06	0.17
Felt cheerful last 2 weeks (WHO)	1843	2.59	1.72	1435	2.64	1.64	0.05	-0.12	0.22
Felt calm & relaxed last 2 weeks (WHO)	1843	2.62	1.68	1435	2.68	1.59	0.06	-0.11	0.22
Felt active & vigorous last 2 weeks (WHO)	1843	2.57	1.64	1435	2.61	1.54	0.04	-0.13	0.21
Woke up fresh & rested last 2 weeks (WHO)	1843	2.58	1.62	1435	2.56	1.53	0.02	-0.14	0.18
Daily life filled with things that interest last 2 weeks (WHO)	1843	2.25	1.76	1435	2.30	1.70	0.12	-0.06	0.29
Felt unable to control important things last month (Cohen)	1843	2.95	1.26	1435	2.92	1.12	-0.06	-0.17	0.05
Felt confident about ability to handle personal problems last month (Cohen)	1843	3.09	1.27	1435	3.05	1.17	0.02	-0.08	0.13
Felt confident things were going your way last month (Cohen)	1843	2.81	1.33	1435	2.89	1.25	0.10	-0.02	0.23
Felt difficulties were piling up could not overcome them last month (Cohen)	1843	2.39	1.28	1435	2.56	1.20	0.10	-0.03	0.23
School attendance (days past week)	299	3.21	2.72	226	2.74	2.72	-0.38	-0.82	0.07

ITT = intention-to-treat effect estimate; HH = household; FIB = fecal indicator bacteria; MPN = most probable number; IHS = inverse hyperbolic spline; WHO = World Health Organization; WVS = World Values Survey. Effects are estimated with models that include controls for randomization blocks based on province and number of villages per cluster. There were 121 clusters in total. The WASH governance perceptions index and the hygiene and behavior index were calculated by rescaling each variable in the index (e.g., satisfaction with water access) so that higher values imply better outcomes, then standardizing relative to the control group, following Kling *et al.* Effects are in standard deviation units.

S4 Table. Intervention effects on all primary outcomes, by province (pre-specified)

		Control			Intervention				CI 95%	
Outcomes	Province	n	Prevalence/Mean	SD	n	Prevalence/Mean	SD	ITT	Lower Bound	Upper Bound
Diarrhea prevalence	Kongo Central	146	17%		163	10%		-0.07	-0.15	0.00
	Kasai	599	40%		679	40%		0.00	-0.09	0.09
	Kasai Central	781	49%		196	59%		0.10	0.00	0.19
	South Kivu	784	41%		724	37%		-0.05	-0.10	0.01
Length-for-age Z-score	Kongo Central	96	-1.59	1.32	93	-1.75	1.67	-0.18	-0.47	0.11
	Kasai	319	-2.05	1.60	328	-1.86	1.60	0.20	-0.05	0.44
	Kasai Central	409	-2.12	1.71	103	-2.58	1.49	-0.42	-0.80	-0.04
	South Kivu	399	-2.49	1.48	395	-2.50	1.50	0.02	-0.16	0.19
WASH institutions index	Kongo Central	18	0.27	0.66	20	0.59	0.57	0.31	-0.06	0.68
	Kasai	43	-0.48	1.10	48	0.21	0.75	0.70	0.21	1.19
	Kasai Central	65	-0.18	0.85	16	0.28	0.61	0.35	0.03	0.68
	South Kivu	59	0.46	0.96	60	0.67	0.78	0.23	-0.25	0.71

ITT = intention-to-treat effect estimate. Effects are estimated with models that include controls for randomization blocks based on province and number of villages per cluster. There were 121 clusters in total. The WASH institutions index was calculated by rescaling each variable in the index (e.g., presence of WASH committee) so that higher values imply better outcomes, then standardizing relative to the control group, following Kling *et al*. Effects are in standard deviation units.

S5 Table. Intervention effects on all primary outcomes, province-by-intervention interaction models

	(1) WASH Intitutions Index - Simple	(2) WASH Intitutions Index - Interaction	(3) Diarrhea - Simple	(4) Diarrhea - Interaction	(5) Height for Age - Simple	(6) Height for Age - Interaction
Intervention	0.404***	0.308*	-0.010	-0.070*	-0.014	-0.172
	(0.125)	(0.167)	(0.022)	(0.036)	(0.070)	(0.134)
Kasai Province # cluster_strata=2	-0.644**	-0.024	0.267***	0.003	-0.085	0.090
	(0.251)	(0.256)	(0.050)	(0.044)	(0.200)	(0.149)
Kasai Province # cluster_strata=3	-0.432*	0.196	0.232***	-0.031	0.222	0.396***
	(0.221)	(0.315)	(0.078)	(0.077)	(0.172)	(0.091)
Kasai Central Province # cluster_strata=2	-0.263	0.185	0.372***	-0.024	-0.347	-0.005
	(0.201)	(0.199)	(0.059)	(0.055)	(0.261)	(0.222)
Kasai Central Province # cluster_strata=3	-0.866**	-0.430	0.335***	-0.036	-0.143	0.106
	(0.410)	(0.411)	(0.063)	(0.059)	(0.168)	(0.140)
Sud-Kivu Province # cluster_strata=2	-0.064	0.042	0.273***	0.035	-0.729***	0.096
	(0.231)	(0.237)	(0.041)	(0.032)	(0.161)	(0.141)
Sud-Kivu Province # cluster_strata=3	0.670**	0.763***	0.268***	0.030	-0.240	0.586***
	(0.262)	(0.223)	(0.042)	(0.028)	(0.164)	(0.142)
Kasai Province		-0.835***		0.226***		-0.359*
		(0.314)		(0.054)		(0.212)
Kasai Central Province		-0.488*		0.339***		-0.323
		(0.272)		(0.050)		(0.223)
Sud-Kivu Province		-0.065		0.224***		-0.912***
		(0.293)		(0.050)		(0.227)
Intervention # Kasai Province		0.391		0.069		0.365**
		(0.293)		(0.058)		(0.182)
Intervention # Kasai Central Province		0.044		0.168***		-0.242
		(0.230)		(0.059)		(0.226)

Intervention # Sud-Kivu Province		-0.076		0.022		0.184
		(0.286)		(0.044)		(0.158)
Child's sex			0.010	0.010	0.134**	0.135**
			(0.014)	(0.014)	(0.062)	(0.062)
Child's age (years)			-0.043***	-0.043***	-0.233***	-0.234***
			(0.004)	(0.004)	(0.033)	(0.033)
Constant	0.272	0.324	0.235***	0.268***	-1.323***	-1.248***
	(0.168)	(0.214)	(0.039)	(0.044)	(0.182)	(0.204)
Observations	329	329	4072	4072	2142	2142
Wald_F		0.790		3.169		3.022
Wald_p		0.502		0.027		0.032

Coefficients and standard errors from linear models of the primary outcomes on intervention group, fixed effects for randomization strata, fixed effects for province, and intervention-by-province interaction terms. Standard errors are clustered by cluster (group of villages). There were 121 clusters in total. The WASH institutions index was calculated by rescaling each variable in the index (e.g., presence of WASH committee) so that higher values imply better outcomes, then standardizing relative to the control group, following Kling *et al*. Effects are in standard deviation units. The Wald F is a test statistic for the null hypothesis that all of the interaction term coefficients are zero. * p<.1, ** p<.05, *** p<.01"

S6 Table. Intervention effects on diarrhea and length-for-age z score, separately by sex (pre-specified)

		Control			Intervention				CI 95%	
Outcomes	Province	n	Prevalence / Mean	SD	n	Prevalence / Mean	SD	ITT	Lower Bound	Upper Bound
Diarrhea	Female	1151	43%		880	38%		-0.01	-0.06	0.04
	Male	1159	41%		882	38%		0.00	-0.06	0.05
Length for age	Female	605	-2.08	1.59	475	-2.19	1.64	-0.07	-0.24	0.10
	Male	618	-2.28	1.61	444	-2.22	1.53	0.06	-0.14	0.26

ITT = intention-to-treat effect estimate. Effects are estimated with models that include controls for randomization blocks based on province and number of villages per cluster. There were 121 clusters in total. The WASH institutions index was calculated by rescaling each variable in the index (e.g., presence of WASH committee) so that higher values imply better outcomes, then standardizing relative to the control group, following Kling *et al.* Effects are in standard deviation units.

S7 Table. Intervention effects on diarrhea and length-for-age z score, sex-by-intervention interaction models

	(1) Diarrhea - Simple	(2) Diarrhea - Interaction	(3) Height for Age - Simple	(4) Height for Age - Interaction
Treatment	-0.010	-0.004	-0.014	0.071
	(0.022)	(0.027)	(0.070)	(0.098)
Child's sex	0.010	0.014	0.134**	0.206**
	(0.014)	(0.018)	(0.062)	(0.084)
Child's age (years)	-0.043***	-0.043***	-0.233***	-0.233***
	(0.004)	(0.004)	(0.033)	(0.033)
Kasai Province # cluster_strata=2	0.267***	0.267***	-0.085	-0.089
	(0.050)	(0.050)	(0.200)	(0.199)
Kasai Province # cluster_strata=3	0.232***	0.232***	0.222	0.213
	(0.078)	(0.078)	(0.172)	(0.174)
Kasai Central Province # cluster_strata=2	0.372***	0.372***	-0.347	-0.351
	(0.059)	(0.058)	(0.261)	(0.260)
Kasai Central Province # cluster_strata=3	0.335***	0.335***	-0.143	-0.150
	(0.063)	(0.063)	(0.168)	(0.167)
Sud-Kivu Province # cluster_strata=2	0.273***	0.273***	-0.729***	-0.737***
	(0.041)	(0.041)	(0.161)	(0.159)
Sud-Kivu Province # cluster_strata=3	0.268***	0.268***	-0.240	-0.242
	(0.042)	(0.042)	(0.164)	(0.163)
Treatment # Female		-0.011		-0.168
		(0.028)		(0.122)
Constant	0.235***	0.233***	-1.323***	-1.354***
	(0.039)	(0.040)	(0.182)	(0.185)
Observations	4072	4072	2142	2142
Wald_F		0.159		1.883
Wald_p		0.691		0.173

Coefficients and standard errors from linear models of the primary outcomes on intervention group, fixed effects for randomization strata, fixed effects for child sex, and intervention-by-sex interaction terms. Standard errors are clustered by cluster (group of villages). There were 121 clusters in total. The WASH institutions index was calculated by rescaling each variable in the index (e.g., presence of WASH committee) so that higher values imply better outcomes, then standardizing relative to the control group, following Kling *et al*. Effects are in standard deviation units. The Wald F is a test statistic for the null hypothesis that all of the interaction term coefficients are zero. * p<.1, ** p<.05, *** p<.01"

S8 Table. Variable definitions

Descriptions of primary and secondary outcomes, including index subcomponents

Category	Outcomes	Definition
Primary Outcome	**Diarrhea**	Prevalence of diarrhea in last 7 days of children under 5 (caregiver reported): 1. Yes to 'diarrhea' , or 2. Yes to 'three or more stools" AND 'watery or soft stool', or 3. Yes to blood in the stool
Primary Outcome	**Length for age**	Length for age z score for children under 5. The sample does not include observations with implausible z-scores.
Primary Outcome	**WASH institutions index**	The index is built from: (1) Presence of water committee; (2) Frequency of meeting; (3) Average amount spent per month for water activities excluding maintenance (inverse hyperbolic sine); (4) Tracks health conditions; (5) Tracks hygiene and sanitation
Subcomponent	Presence of water committee	1 if the village has a water committee 0 Otherwise
Subcomponent	Frequency of meeting	Frequency of committee meeting. The variable ranges from 1 to 6: 1 Weekly 2 Fortnightly 3 Monthly 4 Every 3 months 5 Every 6 or more months 6 No regular schedule, based on needs
Subcomponent	Average amount spent per month for water activities excluding maintenance.	Average amount spent per month for water activities excluding maintenance. An inverse hyperbolic sine transformation has been applied to the variable.
Subcomponent	Tracks health conditions	1 if village keeps track of community health conditions 0 Otherwise
Subcomponent	Tracks hygiene and sanitation	1 if village keeps track of WASH practices 0 Otherwise

Secondary Outcome	**Water governance perception index**	Water governance perception index is constructed based on the following variables: fairness of selection of water governance entity; perception of fair treatment; confidence in managing money of committee; confidence in committee response to breakdowns; confidence in committee management; Satisfaction with water access
Subcomponent	Committee selected fairly	The extent to which the process of choosing committee members was fair The answer choices are: Not fair at all; Somewhat fair; Fair; Fully fair
Subcomponent	Committee treats community fairly	The extent to which the water committee treats you fairly The answer choices are: Very unfair Somewhat unfair Neither fair nor unfair Somewhat fair Very fair
Subcomponent	Committee manages money well	How well the water committee manages money The answer choices are the following: Very badly Badly Neither bad nor good Good Very good
Subcomponent	Comm responds breakdown well	How well the water committee responds to breakdowns The answer choices are the following: Very badly Badly Neither bad nor good Good Very good
Subcomponent	Confidence committee will solve reported issue	Level of confidence that water committee will solve issue respondent brings up The answer choices are the following: Not confident at all Not very confident

		Somewhat confident Very confident
Subcomponent	Satisfaction with water access (1-5)	Level of satisfaction of respondent with your access to water The answer choices are the following: Not satisfied at all Not satisfied Indifferent Satisfied Very satisfied
Secondary Outcome	**Improved water source**	1 if Household's Primary source of drinking water is improved source (JMP definition) and 0 otherwise. ● Improved main drinking source includes Piped into dwelling, Piped into plot, Piped/public tap, Tube well or borehole, Protected dug well, Protected spring, Rainwater, Tanker truck. ● Unimproved main drinking source includes Unprotected dug well, Unprotected spring, Surface water, No other main source
Secondary Outcome	**Time spent collecting water (IHS)**	Each household was asked to list household members who participated on previous day, and the number of trips undertaken by the household. For each trip, the collector was identified and the time spent for the round trip was collected. For each household, we sum time spent for each trip in minutes to get the total time spent collecting water per household. To estimate treatment effects, an inverse hyperbolic sine transformation was applied due to the large number of zeroes.
Secondary Outcome	**HHs pay for water (Dummy)**	1 if household pay for water 0 Otherwise
Secondary Outcome	**Water use expenditure (weekly)**	Estimated household's weekly expenditure for water use (in CDF, Congolese Franc)

Secondary Outcome	**Improved sanitation**	Household uses an improved latrine (JMP definition) ● Improved sanitation includes options such as Flush / Pour Flush: to Piped Sewer System, Flush / Pour Flush: to Septic Tank, Flush / Pour Flush: to Pit Latrine, Ventilated Improved Pit Latrine (VIP), Pit Latrine with Slab, Composting Latrine. ● Unimproved sanitation includes Flush / Pour Flush: to Elsewhere, Flush / Pour Flush: to Don't Know Where, Pit Latrine without Slab / Open Pit, Bucket Latrine, Hanging Toilet, No Facilities (Bush, Open Field, River), Other (specify)
Secondary Outcome	**Hygiene and behaviour index**	Self-reports of: Knowledge: Caregiver knows how and when to wash hands; what causes diarrhea. Sanitation practices: cleanliness of household area and latrine (presence of flies and fecal matter); open defecation; observed indicators of toilet use –worn pathway, presence of water; improvements to latrine; disposes of child feces safely (JMP definition). Water storage practices: has a clean pot for water that is covered.
Subcomponent	Handwashing score (0-10)	Self-report of: Knowledge: Caregiver knows how and when to wash hands. The variables considered are the following: count 1 if respondent mentioned unprompted that they wash hands with soap/ash in critical juncture X (= 0 if they say no or they say it prompted). The counts are then added up to create a score. The considered junctures are the following: After toilet; after washing baby's bottom/changing; after eating; before preparing food; before eating; before feeding/breastfeeding baby; before or after handling children; after taking care of pets or farm animals; after coughing/sneezing; after coming back from the fields.

Subcomponent	Open defecation (%)	1 if no defecation facility used and 0 otherwise
Subcomponent	Self-reported handwashing with soap/ash (%)	1 if respondent washed their hands with soap/ash at least once since previous day 0 Otherwise
Subcomponent	Frequency of latrine cleaning over past 2 weeks	Number of times the latrine has been cleaned in the past 2 weeks
Subcomponent	Water pot is clean and covered (%)	1 if the pot has clean water and is covered 0 Otherwise. This indicator is measured via actual observations
Subcomponent	Water treated for consumption, any method (%)	1 if drinking water stored in household is treated with any product/method for safe consumption 0 otherwise
Secondary Outcome	**Life satisfaction & self-esteem index**	Summary index of 11 questions The 11 questions include the (1) life satisfaction question as defined by the World Values Survey and (2) the 10 questions as defined in Rosenberg's Self-Esteem Scale.
Subcomponent	Life satisfaction (WVS)	All things considered, on a scale of 1 to 10, how satisfied are you with your life as a whole? 1 means completely dissatisfied 10 means completely satisfied
Subcomponent	Feel I am person of worth (Rosenberg)	I feel I am a person of worth, at least on an equal plane with others. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.
Subcomponent	Feel that I have good qualities (Rosenberg)	I feel that I have a number of good qualities. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.
Subcomponent	Inclined to feel I am a failure (Rosenberg)	All in all, I am inclined to feel that I am a failure. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.

Subcomponent	Able to do things as well as oth people (Rosenberg)	I am able to do things as well as most other people. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.
Subcomponent	Feel have not much to be proud of (Rosenberg)	I feel I do not have much to be proud of. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.
Subcomponent	Take a positive attitude towards self (Rosenberg)	I take a positive attitude toward myself. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.
Subcomponent	I am satisfied with myself (Rosenberg)	On the whole, I am satisfied with myself. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.
Subcomponent	Wish could have more respect for myself (Rosenberg)	I wish I could have more respect for myself. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.
Subcomponent	Certainly feel useless at times (Rosenberg)	I certainly feel useless at times. Tell me to what extent you: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about you.
Subcomponent	At times think I am no good at all (Rosenberg)	At times I think I am no good at all. To what extent do respondents: Strongly agree, Agree, Disagree, or Strongly disagree with this statement about them
Secondary Outcome	**Psychological well-being index**	Summary index of 9 questions on well-being in last 2 weeks and stress in last 4 weeks. The questions used here belong to two different sets of questions. The first set is the 5 WHO questions on well-being, and the second set is the Cohen stress scale 4 questions.

		Each of the five WHO statements on well-being refers to how respondents have been feeling over the last two weeks. The answer choices are the following: At no time; Some of the time; Less than half of the time; More than half of the time; Most of the time; All of the time. Questions in the Cohen stress scale, ask the respondent about their feelings and thoughts during THE LAST MONTH. The answer choices are the following: Never; Almost never; Sometimes; Fairly often; Very often.
Subcomponent	Felt cheerful last 2 weeks (WHO)	Over the last two weeks, I have felt cheerful and in good spirits. The answer choices are the following: At no time; Some of the time; Less than half of the time; More than half of the time; Most of the time; All of the time.
Subcomponent	Felt calm & relaxed last 2 weeks (WHO)	Over the last two weeks, I have felt calm and relaxed. The answer choices are the following: At no time; Some of the time; Less than half of the time; More than half of the time; Most of the time; All of the time.
Subcomponent	Felt active & vigorous last 2 weeks (WHO)	Over the last two weeks, I have felt active and vigorous. The answer choices are the following: At no time; Some of the time; Less than half of the time; More than half of the time; Most of the time; All of the time.
Subcomponent	Woke up fresh & rested last 2 weeks (WHO)	Over the last two weeks, I woke up feeling fresh and rested. The answer choices are the following: At no time; Some of the time; Less than half of the time; More than half of the time; Most of the time; All of the time.
Subcomponent	Daily life filled with things that interest last 2 weeks (WHO)	Over the last two weeks, my daily life has been filled with things that interest me. The answer choices are the following: At no time; Some of the time; Less than half

		of the time; More than half of the time; Most of the time; All of the time.
Subcomponent	Felt unable to control important things last month (Cohen)	In the last month, how often have you felt that you were unable to control the important things in your life? The answer choices are the following: Never; Almost never; Sometimes; Fairly often; Very often
Subcomponent	Felt confident about ability to handle personal problems last month (Cohen)	In the last month, how often have you felt confident about your ability to handle your personal problems? The answer choices are the following: Never; Almost never; Sometimes; Fairly often; Very often
Subcomponent	Felt confident things were going your way last month (Cohen)	In the last month, how often have you felt confident that things were going your way? The answer choices are the following: Never; Almost never; Sometimes; Fairly often; Very often
Subcomponent	Felt difficulties were piling up could not overcome them last month (Cohen)	In the last month, how often have you felt difficulties were piling up so high that you could not overcome them? The answer choices are the following: Never; Almost never; Sometimes; Fairly often; Very often
Secondary Outcome	**Handwashing action**	Share of adult household members who were observed washing their hands at any juncture, measured via structured observations. The considered junctures are the following: Before obtaining water from a wide-mouthed storage container; Before cutting or preparing food; Before serving food; Before eating; Before feeding child under 5; Before breastfeeding child; After defecation; After toileting; After cleaning child post- toileting

Secondary Outcome	**Handwashing with soap/ash**	Share of adult household members who were observed washing their hands with soap/ash at any juncture, measured via structured observations. The considered junctures are the following: Before obtaining water from a wide-mouthed storage container; Before cutting or preparing food; Before serving food; Before eating; Before feeding child under 5; Before breastfeeding child; After defecation; After toileting; After cleaning child post- toileting
Secondary Outcome	**School attendance**	Number of days child aged 6 to 18 years old attended school in past week, based on responses to “How many days has this child attended school in the past week?” Children who were not enrolled in school were coded as zero.
Secondary Outcome	**Water point has water**	1 if water point provides water 0 Otherwise (Locked water points during survey are not considered)
Secondary Outcome	**WP Coliforms MPN/100mL**	Water quality at point of collection (village water source) is Most Probable Number (MPN) in 100 mL as defined for Aquagenx CBT EC+TC MPN water quality testing kits and follow WHO standards. For nondetects, we substitute half the lower detection limit.
Secondary Outcome	**HH Coliforms MPN/100mL**	Water quality at point of use (Drinking water stored in the household) is Most Probable Number (MPN) in 100 mL as defined for Aquagenx CBT EC+TC MPN water quality testing kits and follows WHO standards. For nondetects, we substitute half the lower detection limit.
Secondary Outcome	**Weight for age**	Weight for age z score for children under 5.

		The sample does not include observations with implausible z-scores.
Secondary Outcome	**Weight for length**	Weight for length z score for children under 5. The sample does not include observations with implausible z-scores.

S1 Text. CONSORT 2010 checklist of information to include when reporting a randomized trial*

Section/Topic	Item No	Checklist item	Reported in Section/ Paragraph
Title and abstract			
	1a	Identification as a randomized trial in the title	Title
	1b	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts)	Abstract
Introduction			
Background and objectives	2a	Scientific background and explanation of rationale	Introduction
	2b	Specific objectives or hypotheses	Last paragraph, Introduction
Methods			
Trial design	3a	Description of trial design (such as parallel, factorial) including allocation ratio	Study design
	3b	Important changes to methods after trial commencement (such as eligibility criteria), with reasons	NA
Participants	4a	Eligibility criteria for participants	Study design, 9th paragraph
	4b	Settings and locations where the data were collected	Study design, 5th paragraph
Interventions	5	The interventions for each group with sufficient details to allow replication, including how and when they were actually administered	Procedures
Outcomes	6a	Completely defined pre-specified primary and secondary outcome measures, including how and when they were assessed	Outcomes
	6b	Any changes to trial outcomes after the trial commenced, with reasons	NA
Sample size	7a	How sample size was determined	Statistical analysis
	7b	When applicable, explanation of any interim analyses and stopping guidelines	NA
Randomization:			
Sequence generation	8a	Method used to generate the random allocation sequence	Randomization and masking

	8b	Type of randomization; details of any restriction (such as blocking and block size)	Randomization and masking
Allocation concealment mechanism	9	Mechanism used to implement the random allocation sequence (such as sequentially numbered containers), describing any steps taken to conceal the sequence until interventions were assigned	Randomization and masking
Implementation	10	Who generated the random allocation sequence, who enrolled participants, and who assigned participants to interventions	Randomization and masking
Blinding	11a	If done, who was blinded after assignment to interventions (for example, participants, care providers, those assessing outcomes) and how	NA
	11b	If relevant, description of the similarity of interventions	NA
Statistical methods	12a	Statistical methods used to compare groups for primary and secondary outcomes	Statistical analysis
	12b	Methods for additional analyses, such as subgroup analyses and adjusted analyses	Statistical analysis
Results			
Participant flow (a diagram is strongly recommended)	13a	For each group, the numbers of participants who were randomly assigned, received intended treatment, and were analyzed for the primary outcome	Study design
	13b	For each group, losses and exclusions after randomization, together with reasons	Study design
Recruitment	14a	Dates defining the periods of recruitment and follow-up	Results
	14b	Why the trial ended or was stopped	NA
Baseline data	15	A table showing baseline demographic and clinical characteristics for each group	Table 2
Numbers analysed	16	For each group, number of participants (denominator) included in each analysis and whether the analysis was by original assigned groups	Tables 3-4
Outcomes and estimation	17a	For each primary and secondary outcome, results for each group, and the estimated effect size and its precision (such as 95% confidence interval)	Results, Tables 3-4
	17b	For binary outcomes, presentation of both absolute and relative effect sizes is recommended	Tables 3-4

Ancillary analyses	18	Results of any other analyses performed, including subgroup analyses and adjusted analyses, distinguishing pre-specified from exploratory	Results; Supp Info
Harms	19	All important harms or unintended effects in each group (for specific guidance see CONSORT for harms)	NA
Discussion			
Limitations	20	Trial limitations, addressing sources of potential bias, imprecision, and, if relevant, multiplicity of analyses	Discussion, 2nd to last paragraph
Generalizability	21	Generalizability (external validity, applicability) of the trial findings	Discussion
Interpretation	22	Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence	Discussion
Other information			
Registration	23	Registration number and name of trial registry	Study design
Protocol	24	Where the full trial protocol can be accessed, if available	Study design
Funding	25	Sources of funding and other support (such as supply of drugs), role of funders	Abstract

*We strongly recommend reading this statement in conjunction with the CONSORT 2010 Explanation and Elaboration for important clarifications on all the items. If relevant, we also recommend reading CONSORT extensions for cluster randomized trials, non-inferiority and equivalence trials, non-pharmacological treatments, herbal interventions, and pragmatic trials. Additional extensions are forthcoming: for those and for up-to-date references relevant to this checklist, see www.consort-statement.org.